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DETECTION OF FRAUDULENT GCASH ELECTRONIC PAYMENT RECEIPTS USING PATTERN ANALYSIS-BASED IMAGE PROCESSING

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HOLY ANGEL UNIVERSITY**DETECTION OF FRAUDELENT GCASH ELECTRONIC PAYMENT RECEIPTS
USING PATTERN ANALYSIS-BASED IMAGE PROCESSING****Introduction**

An estimated ₱370 billion in government revenue has been lost in the Philippines due to the proliferation of fake and "ghost" receipts used for tax evasion, according to the Bureau of Internal Revenue (BIR) (BIR, 2023; Gatchalian, 2023). This figure is derived from approximately ₱1.3 trillion worth of fraudulent receipts issued by both small and large businesses over the past two decades, with foregone income tax (25%) and value-added tax (12%) comprising the bulk of the loss. The BIR's Run After Fake Transactions (RAFT) program has since intensified enforcement efforts, filing multiple civil and criminal cases against approximately 70 taxpayers for buying, selling, or using fake receipts to substantiate expenses and input VAT claims. In 2024 alone, the BIR collected over ₱4 billion from issuers of ghost receipts, indicating that fraudulent documentation remains a persistent and economically significant problem.

While earlier research on fake receipts has largely focused on paper-based invoices and sales documents, the rapid digitization of payment systems in the Philippines has introduced a new dimension to this problem: electronic receipts from e-wallet platforms such as GCash. These digital confirmations are now widely used as proof of payment in online marketplaces, social commerce, and small-scale business transactions. According to the Cybercrime Investigation and Coordinating Center (CICC), GCash was the top e-wallet used by fraud victims in 2024, with users recording over ₱76 million in losses from various online and consumer fraud schemes. Although these figures cover multiple scam types, they underscore the centrality of electronic receipts and payment confirmations in fraudulent transactions and highlight the vulnerability of digital payment ecosystems.

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The BIR has explicitly identified the proliferation of fraudulent receipts—including digitally generated or electronically transmitted documents—as a significant issue resulting in billions of pesos in lost tax revenue. These enforcement initiatives demonstrate that the government recognizes fraudulent receipts—not only traditional paper invoices but also electronic formats—as a critical threat to revenue integrity.

Despite these efforts, electronic receipt fraud remains underexplored in academic and policy research. GCash has issued public advisories warning the public against AI-generated and edited fake receipts, noting that such fraudulent confirmations are designed to imitate genuine transaction records and are frequently used to deceive merchants into releasing goods without actual fund transfers. The platform has also blocked millions of fraudulent accounts and taken down phishing sites and malicious posts, reflecting the scale of abuse targeting digital payment channels. These advisories recommend verification directly through the transaction history within the GCash app, highlighting the limitations of relying solely on screenshots or shared images as proof of payment.

From a technical and academic perspective, electronic receipt fraud intersects with research in information security, digital forensics, and financial crime. Studies in document and image forensics demonstrate that counterfeit digital documents can be reliably detected by analyzing internal structural patterns—such as layout geometry, text alignment, typographic consistency, and background textures—rather than by superficial visual inspection alone. Techniques such as grayscale normalization, edge detection, texture feature extraction, and supervised classification have been applied to banknotes, passports, certificates, and identity documents, achieving high detection accuracy without directly accessing personal or transactional metadata. However, existing work tends to focus on static

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official documents with highly standardized layouts, whereas e-wallet receipts vary across app versions, device resolutions, and delivery channels (e.g., in-app screens, emails, exported PDFs, screenshots). This variability complicates direct transfer of existing models and underscores the need for specialized pattern-analysis frameworks tailored to e-wallet receipts, particularly in settings like the Philippines where platforms such as GCash play a dominant role in everyday transactions.

At the same time, reliance on backend or API-based verification alone is not always sufficient. Small merchants, especially those operating via social media and messaging apps, often lack the infrastructure or time to cross-check every receipt against provider databases in real time. As a result, decision-making frequently falls back on visual inspection of receipt images, where high-quality forgeries are difficult to detect. This gap highlights the need for a scientific, automated verification system that can operate locally—without continuous access to centralized databases or personal information—to screen electronic receipts for authenticity.

This study proposes the Detection of Fraudulent GCash Electronic Receipts Using Pattern Analysis-Based Image Processing, which leverages visual feature extraction and pattern recognition to differentiate genuine GCash receipts from counterfeit ones. By modeling expected spatial relationships between layout elements (logos, headers, amounts, timestamps, reference codes) and statistical properties of background textures, fonts, and color distributions, the proposed system can flag anomalies indicative of tampering or template misuse. Deployed as a standalone verification tool on merchant devices, this approach supports quick, local screening of receipts before goods or services are released,

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thereby reducing direct financial exposure to fraud and indirectly supporting more accurate transaction reporting for tax and regulatory purposes.

By addressing the vulnerability of electronic receipts—one of the most widely used proofs of payment in digital commerce—this research contributes to strengthening the integrity of the digital financial ecosystem in the Philippines. It extends document forensics and image processing techniques into a domain that is currently under-researched yet economically consequential, and it responds to concrete concerns raised by regulatory agencies, payment providers, and cybercrime authorities regarding the misuse of electronic receipts in fraud schemes. Strengthening the authenticity verification of electronic receipts helps ensure that the benefits of e-wallet adoption—convenience, speed, and financial inclusion—are not undermined by widespread forgery and abuse, thereby safeguarding the integrity of the Philippine digital financial ecosystem.

Review of Related Literature**Advances in Digital Payment Receipt Forgery and Image Manipulation Technologies**

The proliferation of accessible image-editing software and generative artificial intelligence has substantially lowered the technical barrier for producing convincing fraudulent financial documents. Zanardelli et al. (2023), in a comprehensive survey published in *Multimedia Tools and Applications*, established that modern forgery techniques—including copy-move, splicing, and AI-based inpainting—have migrated from specialized forensic concerns into everyday financial fraud, rendering conventional visual inspection unreliable. The findings of AIForge-Doc (2025) corroborated this escalation, reporting that

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digital document forgeries increased by 244% year-over-year in 2024 and, for the first time, surpassed physical counterfeits as the dominant fraud modality—representing a 1,600% growth since 2021—driven primarily by diffusion-model inpainting APIs capable of replacing numerical fields in document photographs within seconds. Fathalla et al. (2023), in the *International Journal of Advanced Computer Science and Applications*, further demonstrated through their DMOBILE-ELA framework that identifying manipulation in digital financial documents requires models sensitive to both coarse structural deviations and fine-grained compression inconsistencies simultaneously.

In the Philippine context, the scale of digital payment adoption amplifies the societal impact of receipt manipulation. The Bangko Sentral ng Pilipinas (BSP, 2024) reported that digital retail payments reached 57.4% of total monthly transaction volume in 2024, up from 52.8% in 2023, confirming the country's successful transition toward a cash-lite economy under its Digital Payments Transformation Roadmap. GCash, operated by Mynt and regulated by the BSP as an electronic money issuer, registered approximately 86 million users in 2023 and commanded roughly 89% of the mobile wallet application market share in the country (BCG & Foxmont Capital Partners, 2024). This massive user base makes GCash receipt fraud both high-frequency and high-impact, reinforcing the urgency of automated, image processing-based detection systems aligned with the methodology proposed in this study.

Fraudulent GCash Electronic Payment Receipts and Societal Impact

Fraudulent GCash payment receipts represent a specific and economically damaging form of digital deception in the Philippines. GCash's official fraud advisory (Mynt, 2024)

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highlighted an escalating wave of AI-generated fake receipts fabricated to perfectly imitate genuine GCash transaction confirmations, urging users to verify payments exclusively through the in-app Transactions tab rather than relying on screenshots. The advisory acknowledged that these forgeries exploit the widespread practice among small merchants of accepting screenshot-based proof of payment. This vulnerability is not unique to the Philippines; Saha et al. (2025), in an academic study on informal e-market fraud, documented cases in Bangladesh where sellers suffered financial losses after delivering goods upon receiving convincing AI-generated fake digital payment receipts—an empirical parallel that underscores the global urgency of the problem. BSP data further confirmed that cyberfraud losses in 2023 were heavily concentrated in account takeover and identity-based schemes, with the Philippine National Police recording a 22% increase in cybercrime cases in the first quarter of 2024 alone (Feedzai, 2024).

The bibliometric study of Priya et al. (2025), examining academic research trends on digital payment fraud across emerging economies, found a marked increase in scholarly attention to receipt and document manipulation fraud post-2020, driven by the widespread adoption of mobile wallets in countries with weaker institutional verification infrastructure. Their analysis reinforced that developing nations with rapidly expanding digital payment ecosystems—such as the Philippines—face disproportionate fraud exposure due to higher reliance on screenshot-based payment proof and limited access to real-time transaction verification tools. This socioeconomic dimension strengthens the case for a technology-driven solution that can be deployed at the point of verification without requiring merchant access to bank backend systems—a core design objective of the present study.

HOLY ANGEL UNIVERSITY**Limitations of Manual Electronic Receipt Verification Methods**

Manual verification of electronic payment receipts is fundamentally constrained by human perceptual limitations and the growing sophistication of forged documents. Nogueira et al. (TechRxiv, 2024), in a comprehensive survey on fraudulent document detection, documented that even trained reviewers exhibit high error rates when evaluating documents manipulated using contemporary editing software, particularly when alterations are confined to numerical fields—such as transaction amounts or reference numbers—within an otherwise authentic-looking template. The survey established that human verifiers lack the ability to detect pixel-level compression artifacts, typographic metric inconsistencies, or statistical noise anomalies that automated systems can quantify. Banirostam et al. (2021), in the Journal of Advances in Computer Engineering and Technology, further noted that manual and rule-based verification methods are inherently reactive—identifying fraud only after patterns are already well-established—whereas machine learning approaches using image features and statistical classifiers enable proactive, real-time detection. GCash itself implicitly acknowledged this gap in its 2024 fraud alert by recommending that users never rely on screenshots as payment proof (Mynt, 2024), a concession that the default verification method used by millions of Filipino merchants is no longer reliably secure against well-crafted forgeries.

Pattern-Based Structural Analysis in Electronic Receipt Authentication

Pattern-based structural analysis exploits the spatial regularity of authenticated document layouts—logo placement, field positions, border geometry, and QR code zones—as forensic benchmarks against which suspect receipts can be compared. Ha and Horak

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(2022), in their study on information extraction from scanned invoice images published in *Signal Processing: Image Communication*, demonstrated that combining layout feature analysis with OCR-based field extraction enables accurate identification of structural deviations in financial documents. Their framework validated that defining document zones based on expected spatial coordinates—effectively a Region of Interest (ROI) strategy—and analyzing each zone independently produces substantially more accurate anomaly detection than holistic document-level analysis. Biten et al. (2023), in the ECCV 2022 Workshops proceedings, further confirmed that layout-aware document processing systems that anchor analysis to specific field locations outperform layout-agnostic approaches in extracting and verifying structured document content, providing a methodological precedent directly applicable to GCash receipt authentication.

The Scaler Topics (2023) technical reference elaborated that ROI-based analysis in computer vision systems substantially improves signal-to-noise ratios by concentrating computational resources on semantically significant image subregions. In GCash receipts, the critical ROIs include the transaction amount field, reference number, sender and recipient names, and timestamp—each of which can be independently analyzed for structural consistency, typographic fidelity, and compression uniformity as part of a layered authentication pipeline. Sun et al. (ICIPAI 2024) reinforced this principle through their multiscale edge similarity study, which showed that authentic documents exhibit characteristic regularity in edge distribution across structural boundaries—regularity that is disrupted by even minor digital manipulation—making ROI-anchored edge analysis one of the most sensitive detection strategies available.

Typography and Layout Consistency in Electronic Payment Receipt Verification

Typographic and layout consistency analysis exploits the strict visual uniformity inherent in system-generated electronic receipts—where a single rendering engine produces all authentic documents with identical font metrics, spacing, and positional alignment—to detect the subtle deviations introduced by manual editing. Nogueira et al. (TechRxiv, 2024) surveyed automated font forgery detection methods and highlighted that the extraction of character-level typographic metrics—height, width, stroke thickness, and inter-character spacing—via OCR, compared against a reference font knowledge base, reliably identifies fraudulent text substitutions. Their review established that typographic analysis is among the most sensitive detection strategies for documents where fraud involves field-level text replacement rather than wholesale fabrication, precisely the fraud pattern most common in GCash receipt manipulation. AIForge-Doc (2025) confirmed that even state-of-the-art generative AI tools produce measurable typographic residuals—font rendering inconsistencies, kerning irregularities, and baseline misalignments—when performing localized number substitution in financial receipts, artifacts that CNN-based feature extraction can systematically identify.

Deep Learning and CNN-Based Electronic Receipt Fraud Detection

Convolutional Neural Networks have become the predominant architecture for image-based forgery detection due to their capacity to automatically learn hierarchical feature representations from raw pixel data without requiring hand-engineered descriptors. Sharma et al. (2023), in a comprehensive analytical review published in *Multimedia Tools and Applications*, concluded that deep learning approaches consistently outperformed

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traditional feature engineering pipelines across diverse document types and manipulation categories. Radhakrishnan et al. (2025), in the Journal of Forensic Sciences, empirically validated this by demonstrating that a CNN-based model combining Multi-Scale Histograms with deep feature extraction achieved 98.5% accuracy, with high precision (97.0%), recall (98.2%), and F1-score (98.1%) on image forgery detection tasks. Li et al. (2025), in their GAN-based deepfake payment detection study, further confirmed that deep learning models trained specifically on payment image datasets achieve detection rates exceeding 95% when distinguishing legitimate from AI-fabricated transaction records.

The 2023 CIISCA conference study comparing seven pre-trained CNN architectures—including MobileNetV2, VGG16, Xception, and ResNet101—using ELA-preprocessed inputs found MobileNetV2 among the top performers for binary authentic-versus-forged document classification on the CASIAV2 benchmark. Tyagi and Yadav (2023), in ForensicNet published in the Journal of Forensic Sciences, demonstrated that CNN architectures employing depth-wise convolutions—central to MobileNetV2's inverted residual design—effectively capture texture-level forensic cues in digital documents across copy-move, splicing, and retouching manipulation types, while the inverted bottleneck technique reduces network parameters without sacrificing accuracy. These converging empirical results from peer-reviewed and conference literature provide strong evidentiary support for MobileNetV2 as the backbone architecture in the proposed GCash receipt fraud detection system.

Multi-Scale and Attention-Based Feature Learning in Receipt Fraud Detection

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Multi-scale feature learning addresses the reality that forgery indicators in financial documents manifest simultaneously at multiple spatial resolutions—from pixel-level compression artifacts to field-level structural anomalies to document-level layout deviations. Cho et al. (MDPI Symmetry, 2025), in their edge-focused deep learning study, introduced Edge Attention and Edge Concatenation layers specifically designed to preserve edge-level features that weaken through standard convolution and pooling operations, demonstrating on the DocTamper and receipt datasets that forgery artifacts are frequently localized at boundaries between manipulated and authentic regions. Li et al. (2023), in IEEE Transactions on Neural Networks and Learning Systems, showed that attentional cross-domain CNN features combining spatial attention mechanisms with multi-scale convolutional hierarchies achieve superior manipulation localization accuracy over single-scale baselines. Sabeena and Abraham (2024), in Multimedia Tools and Applications, confirmed that convolutional block attention mechanisms directing pattern recognition toward the most discriminative image regions significantly improve detection accuracy over non-attended baselines in document forgery tasks.

Banirostan et al. (2021) established through their review in the Journal of Advances in Computer Engineering and Technology that hybrid methods combining structural feature recognition with probabilistic classification consistently outperform single-method approaches for electronic payment fraud detection—a conclusion reinforced by Nogueira et al.'s (TechRxiv, 2024) survey identifying OCR-based feature extraction combined with anomaly detection as among the most effective strategies for financial document authentication. For GCash receipt analysis, multi-scale feature learning is particularly pertinent because forgery indicators exist at the character scale (font rendering), field scale

(layout geometry deviations), and document scale (template structural mismatches) simultaneously. MobileNetV2's inverted residual blocks with varying expansion factors inherently facilitate multi-scale feature aggregation, enabling the proposed system to capture discriminative patterns across this full range of spatial scales without requiring separate parallel processing streams.

Transfer Learning for Data-Scarce Electronic Receipt Fraud Detection

Transfer learning addresses a critical practical constraint in receipt fraud detection: the difficulty of assembling large, balanced, and labeled datasets of fraudulent financial documents, given the sensitivity of financial data and the relative rarity of confirmed fraud specimens. Elaraby et al. (2022), in a study published in Nature Scientific Reports, established that fine-tuning deep neural networks pre-trained on large-scale datasets for low-data target tasks enables competitive performance without requiring domain-specific training data at scale, validating transfer learning as the appropriate strategy for novel financial document classification problems with limited labeled examples. The recurrent few-shot document verification study (arXiv, 2024) addressed generalization in document forgery detection specifically, proposing a Prototypical Network approach that achieved strong performance on unseen document classes—including receipt-type documents in the Findit dataset—demonstrating that transfer-based frameworks can adapt to new document designs without complete retraining, a critical capability given that GCash periodically updates its receipt interface.

The CIISCA 2023 comparative evaluation confirmed MobileNetV2 as a leading transfer learning backbone for document forgery detection, outperforming several heavier

architectures on the CASIAV2 dataset when fine-tuned with ELA-preprocessed inputs. Banirostan et al. (2021) further noted that lightweight pre-trained models offer superior accuracy-to-computation trade-offs for financial fraud detection in deployment contexts where computational resources are constrained—directly relevant to the practical implementation scenario of the present study. For the proposed GCash receipt detection system, transfer learning from MobileNetV2's ImageNet-pre-trained weights provides strong visual feature initialization that reduces the labeled data requirement while maintaining robust discriminative capability for Philippine e-payment receipt pattern analysis.

Comparative Studies on Mobile Wallet Fraud Detection in Southeast Asia

Comparative studies on mobile wallet fraud across Southeast Asia provide important regional context for understanding GCash-specific receipt fraud within the broader digital payment security landscape. Saha et al. (2025), examining informal e-market transactions in Bangladesh involving the bKash mobile wallet, documented structurally identical fraud patterns to those observed with GCash in the Philippines—sellers deceived by AI-generated fake payment receipts accompanied by fraudulent SMS transaction notifications—establishing that fake receipt fraud is a systemic vulnerability of mobile wallet ecosystems rather than a platform-specific anomaly. The Jier Research (2024) overview of digital payment frauds across emerging markets further confirmed this regional pattern, reporting that markets with high mobile wallet adoption and limited automated verification infrastructure—characteristics shared by Bangladesh, Indonesia, Vietnam, and the Philippines—consistently exhibit elevated rates of receipt and screenshot-based payment fraud.

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The European Banking Authority and European Central Bank's joint 2024 Report on Payment Fraud, while focused on EU/EEA markets, provides a benchmark comparison point: electronic payment fraud volumes in structured, heavily regulated markets remained significantly lower than in developing economies, which the report attributed to mandatory strong customer authentication (SCA) requirements and automated real-time fraud monitoring systems. This contrast reinforces the argument that the Philippine digital payment ecosystem—currently transitioning under BSP's Digital Payments Transformation Roadmap and AFASA 2024—requires precisely the kind of automated, image processing-based receipt verification system proposed in this research to reach comparable fraud detection capability without necessitating full backend integration by merchant-side users.

Legal and Policy Context of Electronic Payments and Digital Fraud in the Philippines

The Philippine legal and regulatory framework governing electronic payments and digital fraud has undergone substantial development, establishing the policy environment within which the proposed technological detection system operates. The Cybercrime Prevention Act of 2012 (Republic Act No. 10175) defined computer-related forgery and fraud as criminal offenses, explicitly prohibiting the unauthorized input, alteration, or deletion of computer data with fraudulent intent—provisions directly applicable to the manipulation of GCash electronic payment receipts. The Internet Transactions Act of 2023 (Republic Act No. 11967) strengthened this framework by establishing regulatory oversight of digital commerce and empowering the Department of Trade and Industry to pursue enforcement against online fraud; its Implementing Rules and Regulations, finalized under Joint Administrative Order No. 24-03 in 2024, linked violations involving false or misleading

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digital transaction information to prosecution under the Cybercrime Prevention Act (Lexology, 2024). Complementing these statutory measures, BSP Circular No. 1140 (2022) mandated that all banks and electronic money issuers implement automated real-time fraud monitoring and detection systems, creating a direct regulatory expectation for the type of automated receipt authentication system proposed in this research.

The Anti-Financial Account Scamming Act of 2024 (Republic Act No. 12010, AFASA) represents the most significant recent legislative response to digital financial fraud in the Philippines. Enacted to address the escalating prevalence of financial account scams, AFASA criminalizes money muling, social engineering, and economic sabotage; enhances the BSP's investigative powers over digital financial accounts; and mandates a transition to advanced multi-factor authentication across regulated financial platforms by 2026 (SymphonyAI, 2025). BSP data confirmed that account takeover, identity theft, and social engineering schemes—which frequently involve the use of fabricated payment receipts to deceive counterparties—accounted for 76% of total fraud losses in the Philippine financial system in 2025 (Inquirer Business, 2025). The Cybercrime Investigation and Coordinating Center (CICC) further announced in January 2024 plans to require major social media platforms to deploy artificial intelligence tools to combat online scams, reflecting a government recognition that technological solutions are essential for enforcement at scale (Digital Policy Alert, 2024). The regulatory convergence of RA 10175, RA 11967, RA 12010, and BSP Circular 1140 collectively establishes a strong institutional mandate for automated, image processing-based solutions to electronic receipt fraud detection, situating the present research within national financial security priorities.

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This study aims to develop an automated system for detecting fraudulent GCash electronic payment receipts using pattern analysis-based image processing. It is expected to benefit the following:

Merchants and Online Sellers. The proposed system will help merchants verify the authenticity of GCash electronic receipts, reducing financial losses caused by fraudulent payment confirmations.

Consumers. The study will contribute to safer digital transactions by promoting awareness of electronic receipt forgery and strengthening trust in cashless payment systems.

Financial Technology Companies. The findings may serve as a reference for enhancing receipt verification and fraud prevention mechanisms in digital payment platforms.

Government Agencies. The study may support agencies in strengthening fraud detection initiatives through the application of image processing and pattern recognition techniques for electronic receipt verification.

Future Researchers. This study will provide a reference for future research on electronic receipt authentication, document forensics, and deep learning-based image analysis.

Conceptual Framework

The conceptual framework of this study is divided into two primary components: the development of a GCash electronic receipt fraud detection model and its integration into a mobile-based receipt verification application. The model development phase begins with the collection of a dataset consisting of genuine and fraudulent GCash electronic payment

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receipts. These images serve as the primary input for training and testing the classification model.

Data preprocessing is then performed to standardize the dataset. This includes region of interest (ROI) extraction to isolate key receipt elements such as transaction amount, reference number, sender and receiver details, and timestamp. Additional preprocessing techniques such as grayscale conversion, resizing, and edge detection are applied to enhance relevant visual features and ensure consistency across samples.

Following preprocessing, key structural and visual features of GCash receipts are identified for analysis, including layout structure, typography, spacing, logo placement, QR code regions, and texture consistency. The central processing stage involves training a Convolutional Neural Network (CNN) using transfer learning to develop a classification model capable of distinguishing genuine receipts from manipulated or fraudulent ones based on learned visual patterns.

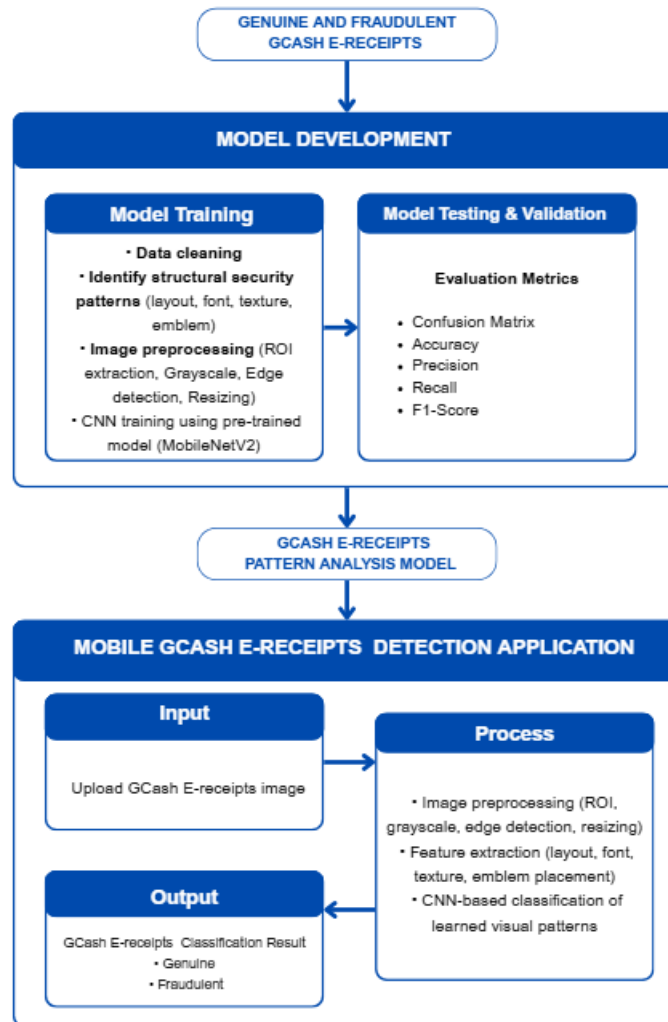
The output phase involves model evaluation using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis to assess classification effectiveness. The validated model is then integrated into a mobile application, allowing users to upload GCash receipt images for verification. The input image undergoes the same preprocessing pipeline before being processed by the trained CNN model, which outputs a classification result indicating whether the receipt is genuine or fraudulent.

Overall, the conceptual framework illustrates the integration of pattern-based image processing and deep learning to form an automated GCash receipt fraud detection system. This system enables real-time verification of electronic payment receipts, supporting

merchants and users in identifying fraudulent transactions and improving trust in digital payment ecosystems.

Figure 1

Conceptual Framework



Objectives of the Study

This study aims to develop a pattern analysis-based system using image processing techniques to detect counterfeit GCash e-receipts. The system will identify visual

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inconsistencies in digital receipts and classify them as genuine or fake through a mobile application that allows scanning and uploading of e-receipts for verification.

Specifically, the study aims to:

- Apply image processing techniques—including edge detection, texture analysis, pattern matching, and layout verification—to extract key visual features of GCash e-receipts, such as transaction layout consistency, font uniformity, QR code structure, timestamp alignment, and overall receipt formatting.
- Develop a CNN-based pattern recognition model that uses the extracted visual features to detect anomalies and discrepancies between genuine and counterfeit GCash e-receipts.
- Test and evaluate the performance of the pattern-based detection model using measurable metrics such as accuracy, precision, recall, F1-score, and false-positive/false-negative rates to ensure reliable counterfeit detection.
- Implement a mobile application that enables users to scan or upload GCash e-receipts for automated verification and instant classification as genuine or counterfeit.
- Assess the feasibility and reliability of the mobile application in real-world scenarios, such as online transactions, peer-to-peer payments, and merchant verification workflows.

Scope and Delimitations

This study focuses on the design, development, and evaluation of a mobile-based pattern analysis system for detecting genuine and fraudulent GCash e-receipts. The system extracts key visual and structural features from e-receipt images, including transaction layout

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consistency, font uniformity, timestamp alignment, spacing, and overall receipt formatting. These features are analyzed to detect anomalies and deviations when compared with standard GCash e-receipt templates, allowing the system to objectively distinguish genuine and fraudulent receipts using image processing techniques rather than manual verification.

The detection model will be implemented through a mobile application developed using the Flutter SDK for Android, with the TensorFlow Lite plugin enabling lightweight, on-device processing. The pattern analysis and classification model will be developed using Python, which will handle image preprocessing, feature extraction, and model training. Users will be able to capture GCash e-receipt images directly using the device upload existing images into the application. The system applies image processing techniques such as edge detection, texture analysis, and pattern matching to analyze submitted receipts and determine whether they are genuine or fraudulent. Offline functionality will be included to allow the system to operate without continuous internet connectivity, making it applicable for peer-to-peer transactions, small merchants, and informal verification settings.

System performance will be evaluated using measurable metrics such as accuracy, precision, recall, F1-score, and false-positive/false-negative rates. These metrics will serve as evidence of the model's reliability in distinguishing genuine and fraudulent GCash e-receipts. However, the system is not expected to achieve perfect accuracy due to variations in screenshot quality, device screen resolution, compression artifacts, and increasingly sophisticated image manipulation techniques.

For testing and evaluation, synthetic and simulated GCash e-receipts will be used. These will be generated based on common transaction formats and known patterns of receipt manipulation observed in digital fraud cases. Standard GCash receipt templates will serve as

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references for genuine samples. No real user data, personal information, or actual financial records will be collected or processed in this study. Furthermore, the research will not involve access to GCash servers, accounts, or any financial databases, ensuring compliance with privacy and data protection standards.

The study is limited to visual and structural pattern analysis. It excludes behavioral and backend verification methods such as account authentication, OTP validation, transaction history verification, and API-based confirmation systems. Other digital payment platforms, including PayMaya, bank transfer receipts, and other e-wallet systems, are also excluded from the scope. Environmental and external factors such as brightness, cropping, compression quality, and image editing tools will be acknowledged as potential influences on detection accuracy but will not be directly modeled in the system's classification process.

The study is not limited to a specific geographic location, as GCash is a nationwide digital payment platform in the Philippines. However, evaluation of the system will be conducted only within controlled academic and local testing environments and will not be deployed in live financial or banking systems. The system is not intended to verify actual financial transactions or resolve payment disputes, but rather to function as a supporting tool for visual fraud detection.

Overall, this research demonstrates the technical feasibility and effectiveness of a pattern-based detection system for identifying genuine and fraudulent GCash e-receipts. It provides a systematic approach to addressing digital receipt fraud while maintaining privacy, ethical considerations, and data security. By focusing on image-based pattern recognition, the study contributes to improving trust and reliability in peer-to-peer digital transactions and may serve as a foundation for future automated receipt verification systems.

HOLY ANGEL UNIVERSITY**Method**

This chapter presents the research methodology of the study, focusing on the procedures and techniques employed to achieve its objectives. It describes the overall research design, explains the selection of participants, details the development and administration of research instruments, and outlines the statistical methods used to analyze the data collected. By providing a structured approach to methodology, this chapter ensures the reliability, validity, and rigor of the study's findings.

Research Design

This study will employ a quantitative research design to systematically collect and analyze numerical data regarding the performance and usability of the GCash e-receipt detection system. A quantitative approach will be selected because the study will require objective measurement of both system functionality and user satisfaction. By collecting measurable data, such as accuracy, precision, recall, F1-score, and structured survey responses, the study will be able to evaluate the system's effectiveness and usability with precision and without subjective bias. This design will also allow for statistical analysis to identify trends, relationships, and areas for improvement, ensuring that the findings will be reliable, valid, and replicable.

Data will be collected from two main sources to address the research objectives. First, model performance metrics will include accuracy, precision, recall, F1-score, and confusion matrix results, which will provide an objective measure of the technical reliability and predictive capability of the system in distinguishing genuine and fraudulent **GCash e-**

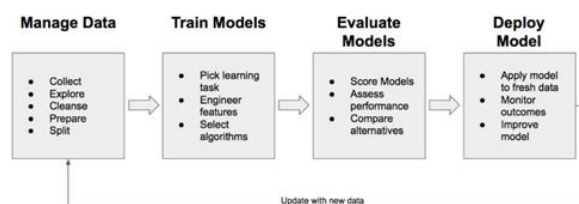
receipts. Second, usability evaluation surveys will be administered through structured questionnaires using a Likert scale to measure user satisfaction, interface ease of use, and overall experience. Together, these quantitative data sources will provide a comprehensive evaluation of both the operational performance and user-centered aspects of the application.

To develop the system systematically, the study will implement the Convolutional Neural Network (CNN) model following a Machine Learning Model Development Lifecycle. This lifecycle will consist of four sequential stages: data management, model training, model evaluation, and deployment. During data management, images of genuine and fraudulent GCash e-receipts will be collected, organized, and pre-processed to ensure data quality. In the model training stage, the CNN architecture will be trained using labeled datasets, while the evaluation stage will assess predictive performance using validation datasets and statistical metrics. Finally, the deployment stage will integrate the trained model into the application for real-time detection and classification. This approach will ensure that the development, validation, and optimization of the machine learning system will be conducted systematically.

Figure 2

Machine Learning Model Development Lifecycle by Chang (2017)

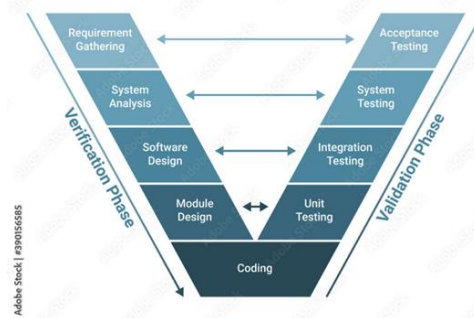
Machine Learning Modeling Cycle



In addition, the V-Model Development Life Cycle will guide the software development process to ensure verification and validation at every stage. System requirements analysis, system design, architectural design, and module development will be conducted to confirm that the application will meet both functional and technical specifications. Subsequently, unit testing, integration testing, system testing, and user acceptance testing will be performed to ensure that the system will satisfy performance standards and user requirements. A detailed discussion of the V-Model implementation will be provided in the Research Procedure section, demonstrating how it will support a disciplined and reliable software development process.

Figure 3

V-Model Development Life Cycle



By combining a quantitative research design with systematic system development procedures, this study will provide a structured and objective framework for evaluating the fraudulent GCash e-receipts detection system. The following sections will detail the data sources, participant selection, technical evaluation procedures, and system development steps, providing a complete and coherent overview of the research methodology.

Sources of Data

This study will utilize multiple sources of data to ensure comprehensive training, validation, and evaluation of the **GCash e-receipt detection system**. These sources include authentic reference images, self-generated and synthetic datasets, and evaluation data from system testing and user assessment.

GCash System-Generated Templates / Standard Reference Data. The image dataset of authentic GCash e-receipts will be based on standard receipt templates and system-generated transaction formats that reflect the official structure of GCash e-receipts. These will serve as references for identifying official visual and structural features of legitimate e-receipts, such as layout configuration, transaction details formatting, font styles, spacing, timestamp placement, and overall receipt structure. Only template-level or anonymized receipt representations will be used, and no personal information, account data, or real financial records will be collected.

Self-generated and synthetic images. To address the limited availability of authentic GCash e-receipt images for model training, the researchers will create their own dataset simulating both standard-pattern and manipulated-pattern e-receipts. Standard-pattern images will follow the typical GCash receipt structure and represent genuine transaction outputs. Manipulated-pattern images will be synthetically produced to mimic fraudulent characteristics, including altered transaction details, misaligned text, inconsistent spacing, distorted layout elements, and abnormal visual artifacts introduced through editing or tampering. These synthetic datasets will be used for model training, validation, and testing (Ahmed et al., 2021; Zhao et al., 2020).

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Evaluation data. Evaluation data will be obtained from the performance outcomes and usability results of the developed mobile application and trained classification model. Quantitative data will include classification metrics such as accuracy, precision, recall, F1-score, and false-positive and false-negative rates. Qualitative data will be collected through structured survey questionnaires administered to participants performing simulated receipt verification tasks. These data will assess the functionality, reliability, and practical applicability of the GCash e-receipt detection system.

Participants

The respondents involved in testing the usability and functionality of the application will be selected using purposive sampling (Palinkas et al., 2015). This method allows the researchers to intentionally choose participants who have direct relevance to the system's intended use and who can provide meaningful insights regarding performance, usability, and practical applicability. Participants will be categorized into three groups: online sellers, small business owners, and field experts.

Online Sellers (Primary Users). This group will represent the primary users of the system. Participants will include individuals engaged in online selling through platforms such as Facebook Marketplace, Shopee, or direct peer-to-peer transactions. These users frequently rely on **GCash e-receipts as proof of payment** and are often exposed to fraudulent or edited receipts. Their evaluation will focus on the practical usability, efficiency, and reliability of the system in verifying transaction authenticity during real-world online sales scenarios.

Small Business Owners (Secondary Users). This group will include micro and small business owners such as sari-sari store owners, food stall operators, and service-based

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vendors who accept GCash payments. They will provide insights into the accessibility, intuitiveness, and overall usability of the application from a practical business perspective. These participants represent secondary beneficiaries who need quick verification tools to reduce the risk of accepting fraudulent payment receipts.

Field Experts. This group will include professionals with relevant expertise, such as cybersecurity specialists, machine learning or computer vision experts, and mobile application developers. Experts will have at least one year of experience in their respective fields and moderate to high technological literacy. Their evaluation will focus on the technical performance, accuracy of the detection model, and system reliability, providing specialized insights that complement the perspectives of end-users.

Research Instrument

The data to be utilized for the development of the application will be collected using the researchers' own computing devices such as laptops and smartphones. Image data will be obtained through simulated GCash e-receipt samples generated from standard receipt formats and controlled transaction templates to ensure consistency in dataset creation. These reference samples will serve as the primary basis for identifying the standard visual and structural features of genuine GCash e-receipts, including layout formatting, transaction details structure, font consistency, spacing, and timestamp placement.

For image acquisition, a mobile camera application will be used to capture screenshots or photographed displays of GCash e-receipts under controlled conditions. This will allow the dataset to include variations in brightness, screen quality, angle, and resolution, ensuring that the system can learn from realistic image scenarios. In addition,

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synthetically generated images will be produced using Canva and similar digital editing tools to simulate fraudulent or manipulated GCash e-receipt samples for training and testing purposes.

To evaluate the application's usability, a standardized survey questionnaire will be employed. This questionnaire will be adapted from the Post-Study System Usability Questionnaire (PSSUQ) developed by Lewis (1992) from IBM. Specifically, Version 3 of the PSSUQ will be used in this study, which consists of sixteen (16) questions measured using a 7-point Likert scale. The questionnaire will be administered using printed module testing forms during system testing sessions to measure the users' perceived satisfaction with the application in terms of system usefulness, information quality, and interface quality.

All instruments used in this study will be grounded on established technical and usability literature. Image data will be reviewed, labeled, and preprocessed to ensure consistency and reliability, while the questionnaire will follow the validated structure of the PSSUQ to maintain measurement accuracy. These instruments directly support the research objectives by providing both technical and user-based evaluation of the developed GCash e-receipt detection system.

Data Collection

An image dataset consisting of both genuine and fraudulent **GCash e-receipts** will be gathered for the development and evaluation of the proposed detection model. Genuine GCash e-receipt samples will be based on standard receipt layouts and system-generated transaction formats that follow the official structure of GCash payment confirmations. These

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will serve as the reference standard for genuine e-receipts and will reflect the required format, layout consistency, and transaction information structure.

Fraudulent GCash e-receipt samples will be created using digital image-editing tools such as Canva and similar applications. These simulated fraudulent samples will be produced by modifying selected visual elements of genuine receipt templates, including transaction details, text alignment, spacing, and layout consistency. Both digital and printed versions (display screenshots and edited images) will be prepared to reflect real-world manipulation scenarios.

Images of both genuine and fraudulent GCash e-receipts will be collected using mobile phones to introduce variation in image quality and device specifications. Each receipt image will be saved from a screen and processed under different conditions such as varying brightness, resolution, and angle to simulate real-world usage. This ensures that the dataset represents different capture conditions commonly encountered in peer-to-peer transactions and online payment verification.

The dataset will include both qualitative and quantitative data. Qualitative data will consist of visual observations of receipt features such as transaction layout, font style, spacing, and formatting consistency. Quantitative data will include measurable information such as the number of images collected per category and standardized image resolution used in dataset preparation. The combination of these data types supports the evaluation of the detection model and ensures that the system can process realistic and varied GCash e-receipt inputs.

Data Analysis

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Data analysis in this study will involve systematically examining and interpreting the collected image datasets of **GCash e-receipts** to evaluate the performance of the MobileNetV2 model for fraudulent receipt detection. It will explain how the data will be processed, summarized, and interpreted to answer the research questions and assess the algorithm's effectiveness in real-world conditions. This foundation will provide a clear framework for subsequent preprocessing and model evaluation steps.

The data analysis will utilize Flutter and Dart as the primary development environment for the mobile application. To implement image processing and deep learning-based pattern analysis, the study will employ the following tools and libraries:

- TensorFlow Lite – For running the MobileNetV2 convolutional neural network model on-device.
- tflite_flutter – Flutter plugin to integrate TensorFlow Lite models with Dart.
- OpenCV for Dart – For preprocessing images, including edge detection, grayscale conversion, and region-based feature extraction.
- image/image.dart – Dart package for image manipulation, resizing, and format conversion.

These tools and methods are appropriate as they allow efficient mobile deployment, real-time image processing, and objective performance evaluation, which is critical for validating the system under real-world conditions. Visual outputs such as confusion matrices, performance graphs, and tables will be generated to interpret results and compare performance across genuine and fraudulent **GCash e-receipts**. Together, these tools and visualizations will support a comprehensive understanding of model performance.

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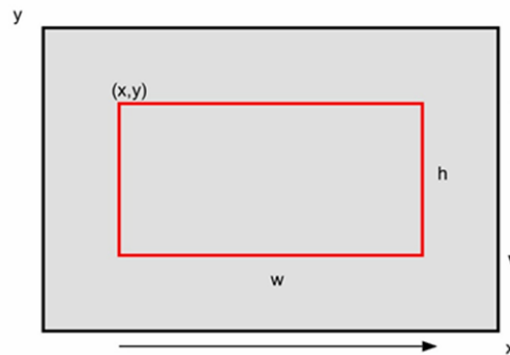
GCash e-receipt images will be collected to include both genuine and fraudulent samples. The images will be captured or obtained using mobile devices to ensure variability across different screen sizes, resolutions, and device types. The receipts will be viewed and captured under varying conditions such as brightness levels and screen display differences to simulate real-world usage scenarios. This approach will provide a diverse dataset that reflects real-world variability in digital transaction verification.

After capturing or collecting the images, the files will be renamed systematically (e.g., “Genuine_01.jpg” to “Genuine_75.jpg” and “Fraudulent_01.jpg” to “Fraud_75.jpg”) for proper organization. Each image will be manually cropped to isolate the receipt area and remove irrelevant background elements such as notifications or borders. This cropping process will produce standardized images suitable for preprocessing. All images will be stored in a designated folder for subsequent analysis. Organizing and standardizing the images in this manner will facilitate consistent preprocessing and accurate model evaluation.

The OpenCV library for Dart will be used to select regions of interest (ROI) for key structural elements of the GCash e-receipt. Using the selectROI() function, critical areas such as transaction details, timestamps, merchant information, and layout sections will be manually selected. The function will return four key values: the x and y coordinates, and the width (w) and height (h) of the selected region. Each detectable feature area will be assigned a unique ROI value. This process ensures that the model focuses on the most relevant structural and visual features for accurate classification.

Figure 4

Selection of Region of Interest

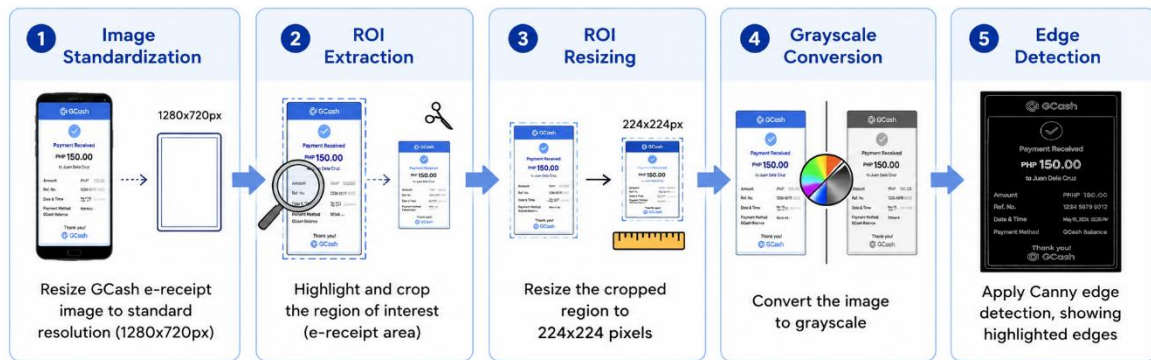


Once the ROI is determined, all images will undergo a preprocessing workflow to ensure uniformity and consistency across the dataset. The preprocessing steps will include:

- *Image Standardization* – All images will be resized to a standard resolution to avoid discrepancies from different mobile camera captures.
- *ROI Extraction* – The regions of interest will be cropped from the standardized images to focus only on relevant security features.
- *ROI Resizing* – Cropped ROI images will be resized to 224×224 pixels, matching the input requirements of the MobileNetV2 model.
- *Grayscale Conversion* – Images may be converted to grayscale to simplify analysis and emphasize structural features over color.
- *Edge Detection* – The Canny Edge Detection algorithm will be applied to highlight structural edges, enhancing the model's ability to recognize key features during classification.

Figure 5

Preprocessing Workflow of Gcash E-receipts



This workflow ensures that the model receives consistent and optimized input, reducing the impact of environmental variations such as screen brightness, device resolution, angle of capture, compression artifacts, and differences between mobile devices. Consequently, the processed dataset will be suitable for reliable model evaluation and performance measurement.

To evaluate the MobileNetV2 model, the researchers will prepare a distinct test dataset of GCash e-receipts and preprocess it using the same procedures applied to the training data. The pre-trained MobileNetV2 model will be loaded using a deep learning framework such as TensorFlow. The researchers will then generate a confusion matrix to visualize the model's performance. The confusion matrix (see Figure 6) will help the researchers analyze the classification results to identify possible areas for improvement and fine-tune the model if necessary, based on insights gained from testing. This evaluation step will directly inform the model's reliability and potential adjustments for deployment.

As noted by Harikrishnan (2019), a confusion matrix for binary classification requires four outcome categories: (1) True Positive (TP), which occurs when the model correctly identifies a fraudulent GCash e-receipt as fraudulent; (2) True Negative (TN), which occurs when the model correctly identifies a genuine GCash e-receipt as genuine; (3) False Positive (FP), which occurs when the model incorrectly identifies a genuine GCash e-receipt as

fraudulent; and (4) False Negative (FN), which occurs when the model incorrectly identifies a fraudulent GCash e-receipt as genuine.

Figure 6

Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

To interpret the results of the confusion matrix, the researchers will define evaluation metrics such as Accuracy, Precision, Recall, and F1 Score. These evaluation metrics will be calculated by comparing the model's predicted labels with the ground truth labels of the test dataset using the values of true positives, true negatives, false positives, and false negatives. This comprehensive testing process will enable the researchers to assess the performance of the MobileNetV2 model and support decisions regarding system deployment or further optimization. These metrics will provide quantitative evidence of the model's effectiveness in differentiating genuine and counterfeit GCash E-receipts.

For binary classification models, such as the proposed GCash E-receipts classification system, accuracy, precision, recall, and the F1 score are key performance metrics used to evaluate machine learning models in classification tasks (Walker). Accuracy will measure the proportion of correct predictions made by the model over the entire dataset. It will be calculated as the ratio of true positives (TP) and true negatives (TN) to the total number of samples, as shown in Equation (1).

$$Accuracy = \frac{TN + TP}{TN + FP + TP + FN}$$

Precision will measure the proportion of true positive predictions among all positive predictions made by the model. It will be calculated as the ratio of true positives (TP) to the sum of true positives (TP) and false positives (FP), as shown in Equation (2).

$$Precision = \frac{TP}{TP + FP}$$

Recall, also known as sensitivity or the true positive rate, will measure the proportion of true positive predictions among all actual positive instances. It will be calculated as the ratio of true positives (TP) to the sum of true positives (TP) and false negatives (FN), as shown in Equation (3).

$$Recall = \frac{TP}{TP + FN}$$

The F1 Score will be used to balance precision and recall. It will be calculated as the harmonic mean of precision and recall and will be useful when a balance between the two metrics is required. Equation (4) shows the formula for computing the F1 Score.

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Together, these evaluation metrics will form a comprehensive framework for understanding model performance and guiding further improvements.

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The data analysis will provide a systematic evaluation of the GCash E-receipts. By using standardized preprocessing, the system will efficiently classify genuine and counterfeit IDs with high accuracy. Performance metrics, including accuracy, precision, recall, F1 score, and confusion matrices, will demonstrate the model's effectiveness. Overall, the results will show that real-time, mobile-based fraudulent detection is feasible and will offer insights for system deployment and improvement.

Research Procedures

This study will adopt the V-Model Development Life Cycle as the guiding framework for the design, development, and evaluation of the Detection of Fraudulent GCash E-receipts Application. The V-Model emphasizes a structured development approach, in which each stage of system construction—from requirement analysis, system and module design, algorithm development, to coding—is rigorously planned and executed. Each development activity is paired with a corresponding verification or validation phase, ensuring that requirements are continuously checked against implementation outcomes, discrepancies are detected early, and system modules meet predefined specifications (Pressman & Maxim, 2020; Somerville, 2022). This alignment enhances reliability, reduces errors, and supports systematic verification at every step, which is critical for an application tasked with distinguishing genuine, and fraudulent GCash E-receipt.

The V-Model is particularly suitable for this research because it integrates testing activities alongside development, allowing each component—such as pattern analysis, image processing, and CNN-based classification—to be individually tested and validated before integration. This methodology ensures that design decisions are verified prior to

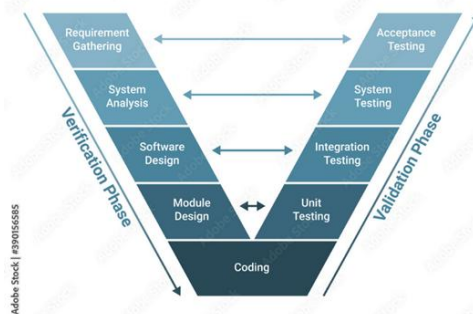
implementation, reducing error propagation and enhancing the robustness of the final system. In addition, the model facilitates the documentation of testing procedures, providing an audit trail for compliance verification and research traceability.

Alternative software development models, such as the Waterfall or Agile methodologies, will not be used in this study. The Waterfall model, while sequential, does not provide continuous verification throughout development, making it less effective for high-accuracy, image-based verification systems. Agile methods, although iterative and flexible, may lead to inconsistent testing and delayed detection of defects, which can compromise system reliability. Previous studies in document verification systems, such as Al-Ghadi, Hassan, and Khalil (2022) and Liao, Zhang, and Chen (2023), have demonstrated that a structured and traceable approach like the V-Model improves detection accuracy and reduces errors, making it more suitable for automated forgery detection applications.

By following the V-Model, this study will ensure that the development of the Detection of Fraudulent GCash E-receipts Application will be methodical, traceable, and aligned with both functional and user-centric requirements, providing a reliable and verifiable system capable of distinguishing genuine and fraudulent GCash e-receipts.

Figure 7

V-Development Life Cycle



HOLY ANGEL UNIVERSITY**Planning Stage**

The Planning Stage will serve as the foundation of the research study. During this stage, the researchers will determine the focus of the study, define the research objectives, identify the problem to be addressed, and establish the overall direction of the project. This stage will also involve identifying the types and quantity of data required for developing and evaluating the GCash e-receipt Verification Application, ensuring that the dataset is sufficient, diverse, and representative for accurate training and testing. In addition, appropriate tools, software, and methods necessary for system development and implementation will be selected.

Furthermore, this stage will include the preparation of the implementation plan, which specifies the goals and targets for each phase of development, testing, and deployment of the application. Project scheduling will be conducted to determine the timeline and logical sequence of activities required to complete the system. This process will support effective planning and execution and will serve as a mechanism for monitoring and controlling project progress. It will ensure that each activity is allocated an appropriate duration and that task dependencies are clearly defined.

A Gantt chart will be used to present the breakdown of tasks across all project phases and to indicate their corresponding schedules and deadlines. The chart will illustrate task timelines, overlaps, and dependencies, providing a structured basis for monitoring and controlling the project schedule (see Appendix A).

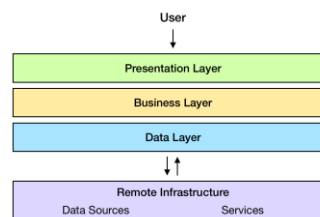
Designing Stage

The Designing Stage will convert the research requirements into a detailed system blueprint. In this stage, the structure and operation of the GCash e-receipt Verification Application will be defined, including how system components will interact and how the algorithm will process input data. The design will focus on planning the system architecture, application workflow, and algorithmic logic to ensure proper functionality, efficiency, and reliability.

The application will be designed as an offline, mobile-based system capable of real-time verification of GCash e-receipts. It will follow a mobile-centric layered architecture, in which all major processing tasks will be executed locally on the mobile device. This architecture will support scalability and maintainability while ensuring efficient performance on devices with limited computational resources. The application will be developed using Flutter and deployed on Android devices for this study, with provisions for future extensibility to other platforms. All operations, including image acquisition, preprocessing, feature extraction, and classification, will be performed locally on the device without reliance on external servers or cloud services. This design will ensure user privacy, reduce response time, and allow the system to function independently of network connectivity.

Figure 8

Mobile-centric layered architecture



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To clearly describe the system structure and behavior, the design will be documented using standard modeling and visualization tools that present the interface layout, workflow, and algorithmic processes of the application. These tools will include flowcharts, use cases, and a UML class diagram.

Flowcharts will provide both textual and graphical representations of the system's processes. Flowcharts will use standardized symbols and connecting lines to demonstrate step-by-step logic. These tools will assist in clarifying and verifying the application's logic prior to development and reduce the likelihood of implementation

errors. Detailed pseudocodes and flowcharts will be included in the appendices (see Appendix B).

Use cases will describe the functional requirements of the system and the interactions between users and the application. They will simulate real-world scenarios, such as capturing a Gcash E-receipt image and receiving a verification result, helping define expected system behavior and responses. Sample use cases will be presented in the appendices (see Appendix C).

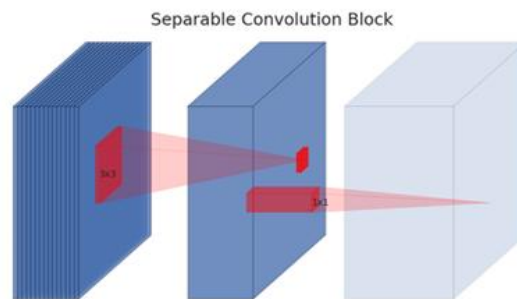
The *UML class diagram* will illustrate the structural design of the system, showing classes, attributes, methods, and their relationships. It will demonstrate how individual modules are organized and interact to form the complete application. The UML class diagram will be included in the appendices (see Appendix D).

The algorithmic core of the application will be implemented using a lightweight convolutional neural network through transfer learning. A pre-trained *MobileNetV2 model* will be used as the base and fine-tuned using a dataset consisting of genuine and fraudulent

GCash e-receipts . This model will be selected due to its balance between computational efficiency and classification accuracy, which is essential for mobile-based verification tasks. The MobileNetV2 architecture will employ depthwise separable convolutions, inverted residual blocks, and linear bottlenecks to reduce model size and computational cost while maintaining performance. These characteristics will enable fast inference and low latency on mobile devices, allowing real-time verification of GCash E-receipts (Howard et al., 2017; Sandler et al., 2018).

Figure 6

Depth wise Separable Convolution in MobileNetV2 (Sandler et al., 2018)



Depthwise separable convolutions will efficiently separate spatial and channel-wise operations, while inverted residuals and linear bottlenecks will improve computational efficiency and maintain classification accuracy during on-device inference. Transfer learning will allow the system to adapt pre-learned visual features to the specific structural and transactional patterns of GCash e-receipts, improving classification accuracy while minimizing training time and data requirements.

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The system workflow will follow a structured sequence of operations during receipt verification. Users will upload images of GCash e-receipts through the mobile application. The images will undergo preprocessing, including resizing, normalization, grayscale conversion, and extraction of regions of interest corresponding to key receipt components such as transaction details, timestamps, and layout structure. Preprocessed images will be analyzed by the MobileNetV2 model to extract relevant features, followed by the classification process in which the system will determine whether the receipt is genuine or fraudulent. The verification result will then be displayed to the user and securely stored locally for future reference. Diagrams illustrating the system architecture, workflow, and structural design will be provided in the appendices to support this discussion.

Development Stage

The Development Stage represents the verification phase of the V-Model, where the GCash e-receipt Verification Application is constructed according to the approved design specifications. This stage focuses on building and implementing the algorithm, coding individual modules, integrating system components, and preparing datasets for testing. The activities are carried out sequentially to ensure that the system meets the functional and non-functional requirements defined during the analysis phase.

Requirement gathering is the first step of the development cycle. The functional and non-functional requirements of the system will be carefully defined, ensuring that the application can correctly distinguish between genuine and fraudulent GCash e-receipts, operate efficiently on mobile devices, and provide an intuitive user experience.

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This phase involves close communication with stakeholders to clarify expected accuracy, system response time, performance specifications, and interface requirements. These requirements will serve as a foundation for subsequent stages, guiding module design, coding, and testing.

Following requirement gathering, system and module design will be carried out to translate requirements into a structured architecture aligned with the mobile-centric layered architecture. This approach ensures that all processing tasks will be executed efficiently on the mobile device while maintaining modularity, scalability, and maintainability.

The High-Level Design (HLD) will outline the layered structure of the system, defining the responsibilities of each layer, data flow between layers, and interactions with external components such as device screens, cameras, and local storage. The layers in the mobile-centric architecture include:

- **Presentation Layer** – Handles user interaction, input acquisition, and display of verification results.
- **Application Logic Layer** – Implements preprocessing, feature extraction, and classification workflows, orchestrating the sequence of operations across modules.
- **Data Layer** – Manages image input/output, local storage of verification history, and temporary datasets used during model inference.

The Low-Level Design (LLD) will detail individual modules within these layers, specifying input/output formats, internal logic, preprocessing steps, feature extraction methods, and classification workflows. Each module will be developed to function independently yet integrate seamlessly within the layered structure, facilitating implementation and unit testing. Modules to be designed include:

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- **Image Upload Module** – Allows uploading images from local storage.
- **Preprocessing Module** – Performs resizing, normalization, grayscale conversion, and region-of-interest (ROI) extraction.
- **Feature Extraction Module** – Implements convolutional neural network layers for extracting distinguishing features.
- **Classification Module** – Uses the MobileNetV2 model with transfer learning to classify receipts.
- **Result Display Module** – Presents verification outcomes to the user and stores verification history locally.

This layered, mobile-centric design will provide a clear roadmap for coding, integration, and future testing, ensuring that each module will operate efficiently on the mobile device while maintaining the modular structure required for maintainability and scalability. By mapping each module to its corresponding layer, the architecture guarantees that real-time processing, offline functionality, and efficient resource use are prioritized for mobile deployment.

Before coding the algorithm, dataset preparation will be conducted. A dataset comprising genuine and fraudulent GCash e-receipt images will be collected under varying lighting conditions and from multiple mobile devices to ensure diversity and improve model generalization.

The images will be cleaned, annotated, and formatted for model consumption. Subsequently, the dataset will be split into (Brownlee, J. 2020):

- **Training Set (80%)** – For model learning and adaptation.

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- **Validation Set (10%)** – To monitor training progress and prevent overfitting.
- **Test Set (10%)** – To evaluate final model performance during the Implementation Stage.

Dataset preparation at this stage ensures that all modules can be developed and tested using real, representative data, which is critical for accurate and reliable GCash e-receipt verification.

Following system design and dataset preparation, coding will commence. Each module will be programmed sequentially, integrating the algorithm with the application interface.

Key activities include:

- Loading and preprocessing images from the dataset.
- Implementing the preprocessing pipeline for normalization, resizing, grayscale conversion, and ROI extraction.
- Integrating the MobileNetV2 model with custom classification layers for GCash e-receipt verification.
- Developing application interfaces for Upload, Home, History, and Learn More pages.
- Connecting modules to ensure smooth data flow and functionality.

By coding the modules in this sequence, the system architecture defined during the design phase will be realized effectively, enabling subsequent integration and testing.

After individual modules are developed, they will be integrated into a cohesive system. Integration ensures that the image capture, preprocessing, feature extraction, classification, and result display modules interact correctly. This step allows early detection of data flow issues and ensures that the application operates as a complete system.

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The Development Stage establishes the foundation for a reliable and efficient Detection of Fraudulent GCash e-receipt Application. By sequentially conducting requirement analysis, system and module design, dataset preparation, coding, and integration, the system will be ready for the Implementation Stage, where algorithm testing, performance evaluation, and user acceptance testing will validate its functionality. Each activity in this stage ensures that the application aligns with its design specifications and can perform robust real-time verification on mobile devices.

Implementation Stage

The Implementation Stage represents the validation side of the V-Model, focusing on testing the algorithm, conducting experiments, and evaluating the performance of the GCash e-receipt Verification Application using the prepared dataset. This stage ensures that the system functions as intended, meets the design specifications, and satisfies end-user requirements.

During this stage, the application will undergo several testing methodologies, including accuracy testing, unit testing, integration testing, system testing, and user acceptance testing. Each method is designed to systematically evaluate the functionality of individual components, the interaction between integrated modules, and the overall performance of the application.

Accuracy Testing

Accuracy testing will evaluate the ability of the application to correctly classify genuine and fraudulent GCash e-receipts. This phase will use representative datasets composed of authentic receipt templates and simulated fraudulent samples generated based

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on common manipulation patterns and digital fraud characteristics. To account for hardware variability, testing will be conducted on multiple mobile devices, including Samsung S24 Ultra, Poco C85, and Realme C35, which vary in camera quality and processing capabilities. Images will be tested under different lighting conditions, screen brightness levels, and capture angles to simulate real-world scenarios. Accuracy will be quantified as the percentage of correctly classified receipts relative to the total number of tests conducted, providing a measure of the effectiveness of the MobileNetV2 classification model and preprocessing pipeline in distinguishing genuine from fraudulent GCash e-receipts (Appendix U).

Unit Testing

Unit testing will verify the functionality of individual components within the application, including image upload, preprocessing, feature extraction, classification, and result display. Each module will be tested independently to ensure proper operation before integration. The development team will conduct these tests in a Python-based test-driven environment, using manual testing and automated scripts to simulate various scenarios. Identified issues will be documented in a testing module checklist and resolved prior to integration, ensuring seamless functionality of each module.

Integration Testing

Following unit testing, all modules will be integrated to form the complete application. Integration testing will verify that modules interact correctly and perform their assigned functions without interfering with other components. Different input scenarios, including variations in brightness, screen resolution, and image orientation, will be tested to detect conflicts, inconsistencies, or data flow errors. Any issues discovered will be

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documented and addressed through debugging and refinement, preparing the system for full-scale system testing.

System Testing

System testing will evaluate the overall functionality, reliability, and performance of the fully integrated application. This phase will simulate real-world usage scenarios, including sequential user interactions and batch verification of multiple GCash e-receipts. Parameters such as classification accuracy, response time, latency, and robustness against varying conditions will be monitored to ensure the system meets design specifications. The research team, together with field experts, will oversee the testing process, documenting any issues, recommendations, and adjustments required to maintain system quality (Appendix V).

User Acceptance Testing (UAT)

User Acceptance Testing will involve representatives from the target user group, including individuals who frequently engage in GCash-based transactions, such as online sellers, small business owners, and digital payment users. Participants will capture or upload GCash e-receipt images and review verification results. Feedback will be collected using standardized instruments, such as the Post-Study System Usability Questionnaire (PSSUQ), to assess usability, interface intuitiveness, and practical effectiveness. Insights gathered from UAT will inform refinements to optimize the user experience and ensure that the application is ready for deployment.

The Implementation Stage validates the GCash e-receipt Verification Application through systematic evaluation and testing. Accuracy testing ensures reliable classification of genuine and fraudulent receipts under varied conditions. Unit and integration testing confirm correct functionality and seamless interaction between modules. System testing evaluates

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overall performance and reliability, while User Acceptance Testing ensures that the application meets end-user expectations. Through this rigorous validation process, the application is verified to function effectively in real-world conditions, fulfilling its design objectives and providing a practical, mobile-based solution for GCash e-receipt verification.

Ethical Considerations

The following ethical guidelines will be strictly followed during the conduct of this study and assessment of its results to ensure the protection of participants, the integrity of data, and compliance with national laws.

Voluntary Participation of the Evaluators

Participation in the research will be entirely voluntary, and all respondents, including online sellers, small business owners, and field experts, will be informed that they can withdraw at any time without penalty or negative consequences. Participants will also be assured that they can refuse to answer any question or opt out of testing while maintaining the confidentiality of the information they provide.

Informed Consent for the Participants of the Study

Before participation, all respondents will be provided with printed informed consent forms that will include detailed information about the study, such as its title and objectives, research methods and procedures, participant roles and responsibilities, data collection processes, criteria for participant selection, anticipated benefits, and measures for confidentiality and data protection. Only those who voluntarily sign the informed consent forms will be included in the study, ensuring that participation will be fully informed and autonomous.

HOLY ANGEL UNIVERSITY**Confidentiality and Data Protection**

All personal information and responses collected from participants will be treated as strictly confidential. No identifying information, such as names or actual financial details from GCash e-receipts, will be stored or included in any research output. Data will be reported only in aggregate form to maintain anonymity. All digital and physical documents containing participant information will be stored in secured, password-protected folders and locked storage systems. Following the conclusion of the study, all records containing identifiable data and survey responses will be permanently deleted. These procedures will comply with the Data Privacy Act of 2012 (Republic Act No. 10173), which mandates lawful processing and protection of personal data.

Ethical Use of Data in Image Datasets

The study will involve the use of synthetic and self-generated **GCash e-receipt images** to simulate genuine and fraudulent transaction records. No actual user accounts, personal financial information, or real transaction data will be accessed, ensuring compliance with privacy laws. This approach respects user confidentiality and prevents potential misuse of sensitive financial information.

Data Integrity and Research Honesty

The study will adhere to principles of accuracy, transparency, and integrity in data collection, analysis, and reporting. All datasets, including self-generated and synthetic GCash e-receipt images, will be documented to maintain replicability and verifiability of results. The research will also strictly follow academic integrity standards, avoiding plagiarism and giving proper citation to all referenced works, in accordance with Republic Act No. 7722 – Higher Education Act of 1994 and institutional guidelines of Holy Angel University.

HOLY ANGEL UNIVERSITY**Minimization of Harm and Social Responsibility**

The research design will ensure that participants and the public are not exposed to risk or harm, whether physical or psychological. The use of synthetic datasets ensures that no real financial or personal data are compromised. Additionally, the study will contribute to social welfare by developing a tool that helps detect fraudulent **GCash e-receipts**, reduces deception in peer-to-peer transactions, and supports safer digital payment practices in accordance with ethical research standards.

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Appendix A

Work Breakdown Structure

Phase	WBS Code	Task	Duration (days)
Phase I: Project Planning, Analysis, and Design	1.1	Brainstorming	14
	1.2	Proposal Preparation	7
	1.3	Proposal Creation	35

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	1.4	Proposal Preparation	5
	1.5	Proposal Revision	10
	1.6	Proposal Submission	2
Phase II: Project Coding and Implementation	2.1	Development Preparation	12
	2.2	Design and Planning	12
	2.3	Data and Requirements Gathering	60
	2.4	Data Modelling and Training	70
	2.5	Application Development	21
	2.6	Module Integration	18
	2.7	Troubleshooting and Debugging	14
Phase III: Project Testing, Deployment, and Maintenance	3.1	Functional Testing	22
	3.1.1	Unit Testing	3
	3.1.2	Integration Testing	3
	3.1.3	System Testing	5
	3.1.4	User Acceptance Testing (UAT)	10
	3.2	Application Documentation	5
Phase IV: Data Analysis and Reports	4.1	Data Analysis	10
	4.2	Report Writing	15

Gantt Chart

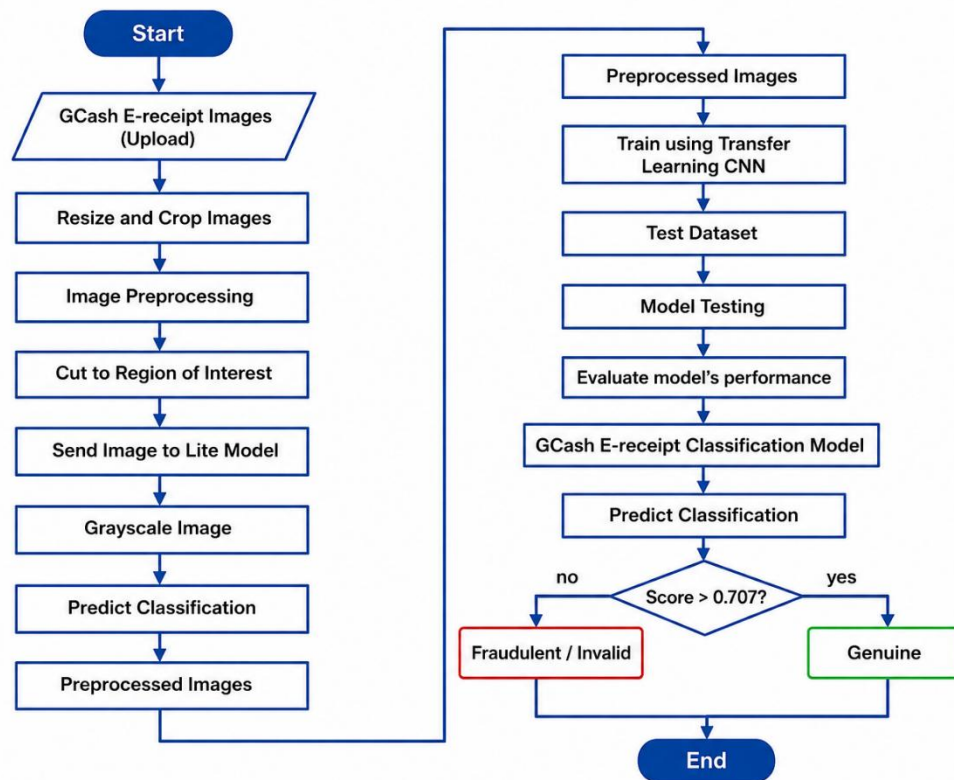
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WBS	Task Name	Start Date	End Date	Duration (days)	Dec '25	Jan '26	Feb '26	Mar '26	Apr '26	May '26	Jun '26	Jul '26	Aug '26	Sep '26
					D W Y	D W Y	D W Y	D W Y	D W Y	D W Y	D W Y	D W Y	D W Y	D W Y
Phase I: Project Planning, Analysis, and Design														
1.1	Brainstorming	12/01/25	12/18/25	18 days										
1.2	Proposal Preparation	01/10/26	01/16/26	7 days										
1.3	Proposal Creation	01/16/26	02/22/26	38 days										
1.4	Proposal Revision	02/23/26	03/03/26	10 days										
1.5	Proposal Submission	03/04/26	03/05/26	2 days										
Phase II: Project Coding and Implementation														
2.1	Development Preparation	03/06/26	03/17/26	12 days										
2.2	Design and Planning	03/18/26	03/29/26	12 days										
2.3	Data & Requirements	03/30/26	05/28/26	60 days										
2.4	Data Modelling & Training	05/29/26	07/28/26	61 days										
2.5	Application Development	07/29/26	08/18/26	21 days										
2.6	Module Integration	08/19/26	09/04/26	17 days										
2.7	Troubleshooting & Debugging	08/19/26	09/04/26	17 days										
Phase III: Project Testing, Deployment, and Maintenance														
3.1	Functional Testing	08/25/26	09/04/26	11 days										
3.1.1	Unit Testing	08/25/26	08/27/26	3 days										
3.1.2	Integration Testing	08/28/26	08/30/26	3 days										
3.1.3	System Testing	08/31/26	09/03/26	4 days										
3.1.4	User Acceptance	09/04/26	09/04/26	1 day										
3.2	Application Documentation	09/01/26	09/04/26	4 days										
Phase IV: Data Analysis and Reports														
4.1	Data Analysis	(TBD)	(TBD)	10 days										
4.2	Report Writing	(TBD)	(TBD)	15 days										

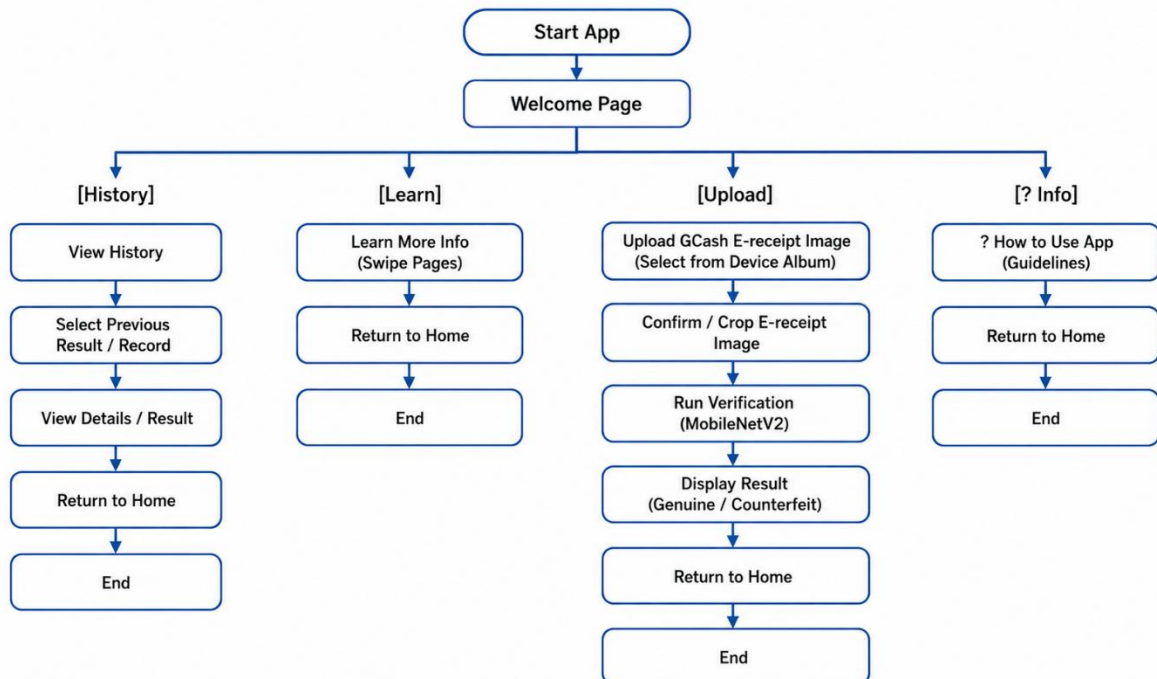
Appendix B

Flowchart for Algorithm

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Flowchart for Application



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Appendix C

Use Case

Use Case ID: UC-1

Use Case Name: General Use Case

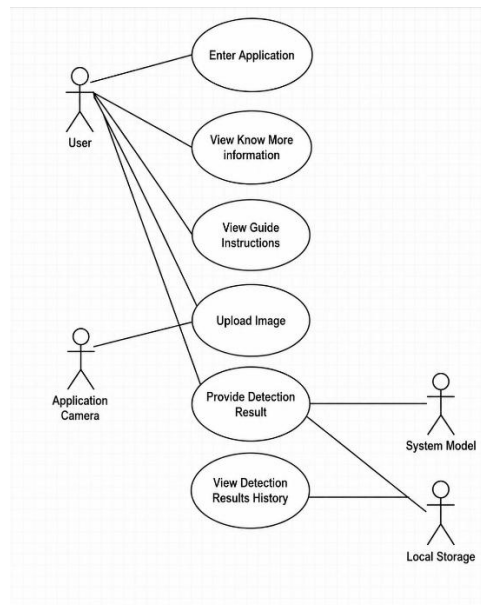
Actor/s: User, Application Camera, System Model, Local Storage

Description: This diagram summarizes the different interactions that can be done within the system.

Precondition: None

Postcondition: None

I. General Use Case



Use Case ID: UC-2

Use Case Name: Enter Application

Actor/s: User

Description: The user initiates interaction by opening the application on a mobile device.

Precondition: The application must be downloaded on the phone.

Postcondition:

1. The user gains access to the application.
2. The user will know how to navigate and use the application.

Actor Action	System Response
User opens the application through an Android mobile device.	The application opens and launches the landing page.

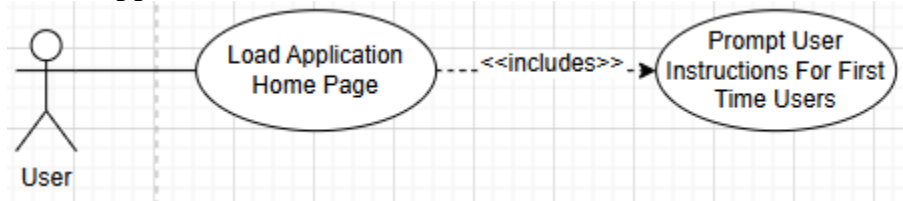
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The system loads the onboarding page.

The user can swipe to view more guidelines, select skip to view the last guidelines, or click the "Tap to Start" button to exit the onboarding page.

The system redirects the user to the home page once the "Tap to Start" button is selected.

II. Enter Application



Use Case ID: UC-3

Use Case Name: View Learn More Information

Actor/s: User

Description: The user selects the "Learn More" button to view additional information.

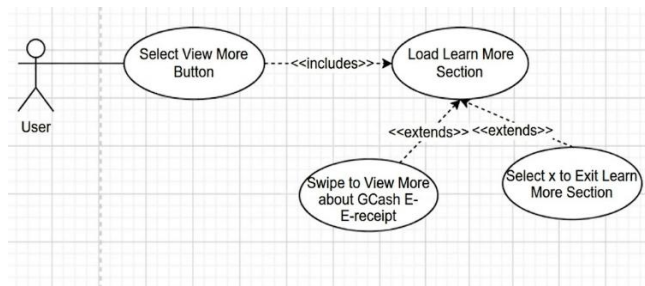
Precondition: The application's homepage should be on screen.

Postcondition: The user will know additional information about the Philippine

Actor Action	System Response
User clicks on the "Learn More" button.	The application opens the "Learn More" section.
	The user swipes to view the different information pages.
User selects the exit button.	The system closes the "Learn More" section and returns to the homepage.

III. View Learn More

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Use Case ID: UC-4

Use Case Name: View Guide Instructions

Actor/s: User

Description: The system loads the guidelines again once the user selects the '?' button.

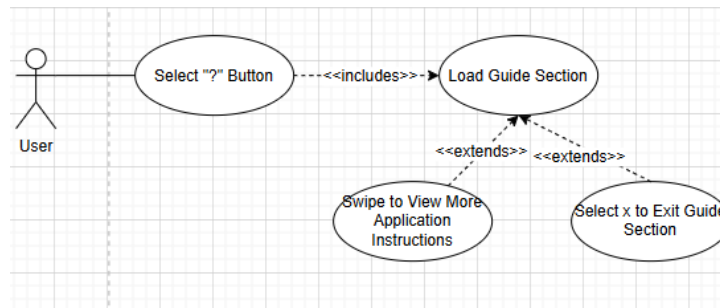
Precondition: The application's homepage should be on screen.

Postcondition:

1. The user will know how to navigate and use the application.
2. The application returns to the homepage.

Actor Action	System Response
User selects the '?' button.	The system loads the guide section.
	The user can swipe to view more guidelines, select skip to view the last guidelines, or click the "Let's Go" button to exit the onboarding page.
	The system redirects the user to the home page once the "Let's Go" button is selected.

IV. View Guide Instructions



Use Case ID: UC-5

Use Case Name: Upload Image

Actor/s: User

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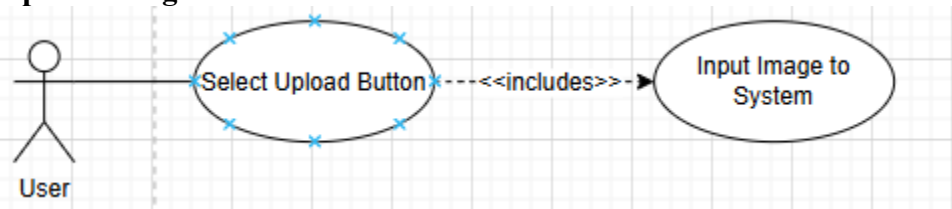
Description: An image in the user's gallery is uploaded to the system as input.

Precondition: Users should have a image on their mobile devices.

Postcondition: An image is uploaded as an input to the system.

Actor Action	System Response
User selects the upload button.	The system launches the cropping page.
User crops the image and selects the check button.	The image is used as input to the system.

V. Upload Image



Use Case ID: UC-6

Use Case Name: Capture Image

Actor/s: User, Application Camera

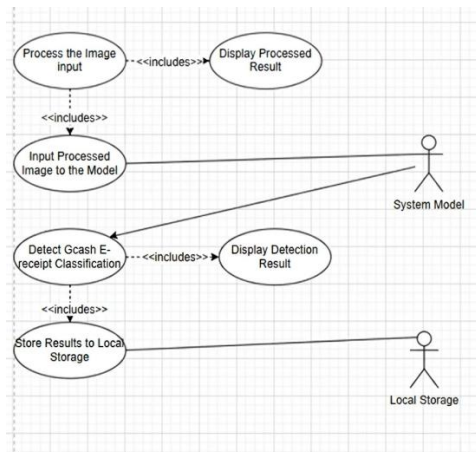
Description: The image that will be used as input is captured via the application.

Precondition: User's phone must have a camera, and the application must have access to it.

Postcondition: An image is captured and used as input to the system.

Actor Action	System Response
User selects the capture button.	The system launches the application camera.
User selects capturing method (manual or auto).	The application camera accesses the device camera.
User selects cropping method (manual or auto).	The system displays the captured image and allows cropping.
User selects the check button.	The image is used as input to the system.

VI. Provide Detection Result



Use Case ID: UC-7

Use Case Name: View Detection Results History

Actor/s: User, System Database

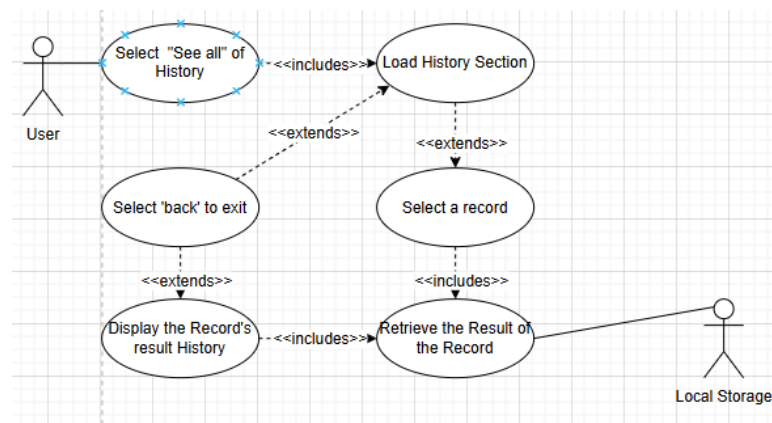
Description: The system displays the detection result history of the user.

Precondition: None

Postcondition: Users can revisit their previous detection results.

Actor Action	System Response
User selects the 'see all' option.	The system loads the history section, showing transaction numbers.
User scrolls and selects a record.	The system retrieves the record from local storage.
	The system displays the chosen record results.
User clicks the back arrow.	The system redirects the user to the homepage.

VII. View Detection Result History



Appendix D

UML Class Diagram

